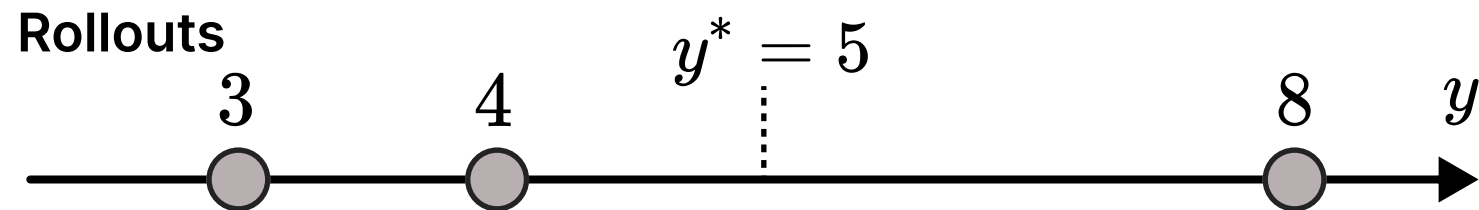


Pointwise Reward (e.g., MSE)

Rewards Each Rollout Alone



How close is this rollout to target?

Rollout 3

$$(3 - 5)^2 = 4$$

Less accurate

Up-weight ↑

Rollout 4

$$(4 - 5)^2 = 1$$

Most accurate

Up-weight ↑

Rollout 8

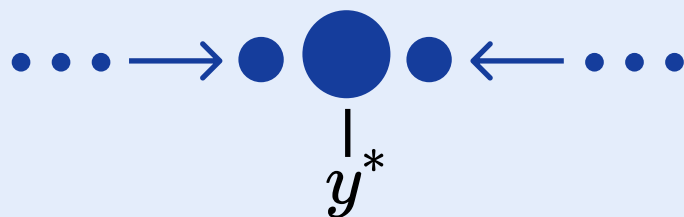
$$(8 - 5)^2 = 9$$

Largest error

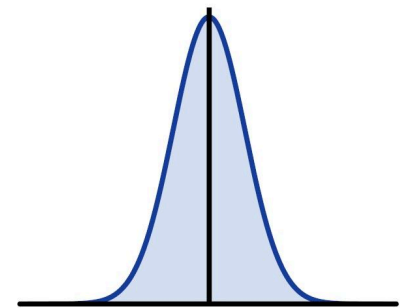
Down-weight ↓

Resulting Update

Conditional mean seeking



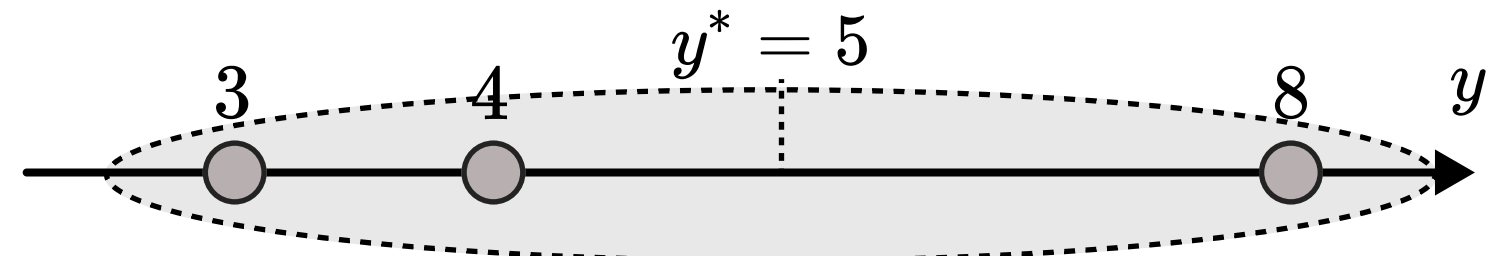
Learned Distribution



⚠ *Narrow & Uncalibrated*

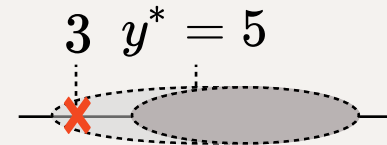
Distribution-Aware Reward (DAR)

Rewards How Each Rollout Changes the Distribution



How does this rollout improve dispersion around the target?

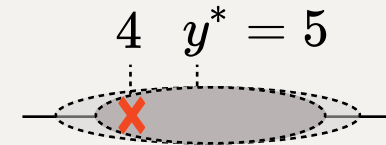
Rollout 3



Hurts dispersion

Down-weight ↓

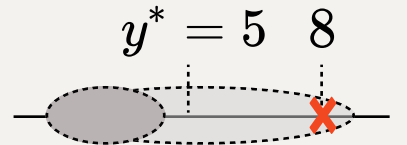
Rollout 4



Improves center

Up-weight ↑

Rollout 8

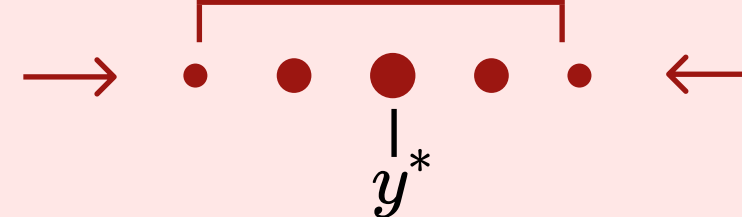


Expands coverage

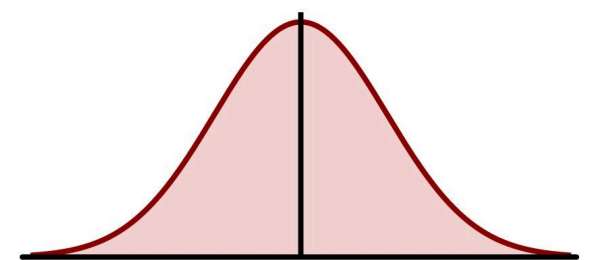
Up-weight ↑

Resulting Update

*Predictive distribution shaping
useful spread*



Learned Distribution



✓ *Well-Spread & Calibrated*